2026 Incheon RoboCup

PISHNAM

RESCUE LINE

We spent 11 months to develop this robot

Arvin MirzaAghayan

Software Developer

Expriences :

- **IranOpen 2023** — 1st Place - Rescue Line Primary
- **IranOpen 2024** — 2nd Place - Rescue Line
- **IranOpen 2025** — 2nd Place - Rescue Line
- **QFR RCJ 2026** — 1st Place - Rescue Line

Arvin is the head of our programming team. He has gained valuable experience and has played an essential role in our team's success. Arvin is a programmer with strong expertise in image processing, highly proficient in Python and skilled in C++. He and Reza work closely on software and algorithms, forming the core of the programming team. With his experience, Arvin plays a key role in problem-solving and helps overcome programming challenges reliably.

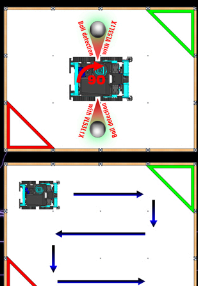
QFR RCJ 2026

Algorithms

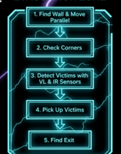
After the robot enters the evacuation zone, it uses two VL53L1X sensors mounted on the left and right sides to scan for victims. It first searches both sides of the evacuation zone (left boundary and right boundary) to quickly detect any victims near the edges. After completing the side scan, the robot performs a complete search of the evacuation zone, continuing to use both the left and right VL53L1X sensors to cover the whole area

In the evacuation zone, the robot first locates the furthest wall, turns to face it, and moves forward to scan the surroundings. It then follows the wall, traveling parallel to it and making 90° turns at corners to systematically cover the area. Using the VL and IR sensors on the APDS board, it detects victims between or near its grippers and picks them up. After rescuing the required victims, the robot continues wallfollowing with VL, IR, and the VL53L1X sensor to search for the exit. Silver tape on the floor indicates the path back toward the evacuation zone, and a black line marks the exit; when this line is detected, linefollowing is activated. This strategy provides reliable victim detection, full coverage of the zone, and efficient evacuation.

Searching for victims :



EVACUATION ZONE




```
Algorithm RobotEvacuation
  WHILE robot not at exit DO
    move_along_wall()

    IF victim_detected THEN
      stop()
      grab_victim()
      confirm_pickup()
    END IF

    IF black_line_detected THEN
      follow_line_to_exit()
    END IF

  END WHILE
End Algorithm
```

CHALLENGE SUMMARY

Team IRAN - Robo Cup Junior World Championship

Engineering Improvments And Performance Validation

IRAN

CHALLANGE	ISSUE	% BEFORE FIX	FIX	% ADDED	Succes Rate
Speed Bumps On Ramps	Getting Stuck	58%	Added Groves To The Tires	+26% 58% → 84%	84%
Obstacle Avoidance	Algorithm Miss Reads	46%	Revised ARC Algorithm	+46% 46% → 92%	92%
Victim Detection	Grip Faliure	47%	Gripper Redesign	+47% 47% → 94%	94%
Intersection With Debris	Detection Failure With Debris	44%	Added Pusher Blade	+52% 44% → 96%	96%
Evacuation Zone	Poor 2025 Performance	38%	Full Routine Rebuild	+38% 38% → 76%	76%

Average Improvment +41.8% — Overall Succes Rate After Fixes 89.6%

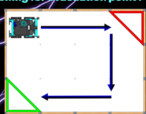
BUILT ON CHALLENGES DRIVEN BY PASSION

WE REPRESENT IRAN.

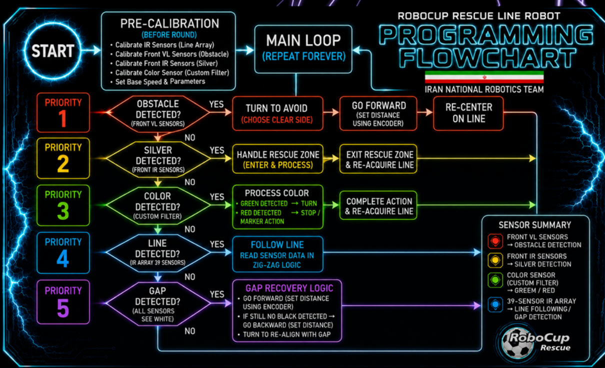
RoboCup Junior

When a victim is detected, the robot switches from searching to locating the evacuation point(s). Once an evacuation point is found, the robot drops off the victim to complete the rescue. After drop-off, the robot uses changes in distance readings from the VL53L1X sensors to determine where the exit is, then leaves the evacuation zone

Searching for evacuation point :



PROGRAMMING



Platform IO :

PlatformIO was chosen because it provides better project organization, easier library management, and a more efficient workflow compared to the Arduino IDE. Using C++ also allows us to write fast and reliable code for real-time robot control.

Reza Bagheri

Software Developer

- **IranOpen 2025** — 1st Place - Rescue Line Primary
- **QFR RCJ 2026** — 1st Place - Rescue Line

Reza Bagheri is the programmer of our team, with exceptional talent in programming and algorithm design. His expertise and problem-solving ability make him a key pillar of our team and a major contributor to the quality of our projects. In addition, he is highly proficient in C++, which further strengthens his role in developing efficient and reliable software for the team.

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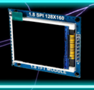
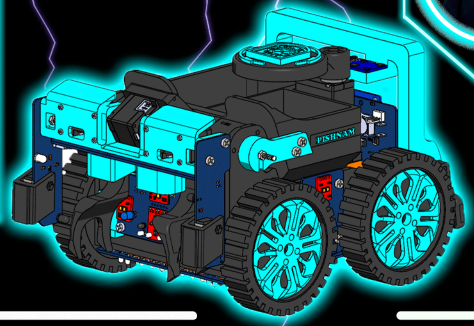
Programming Innovation



Our robot features an on-board menu system with a TFT display and navigation buttons for fast, user-friendly configuration. It enables quick IR sensor calibration and parameter adjustments directly in the field, eliminating the need to reconnect or recompile and improving deployment efficiency.



ELECTRONIC

Amin Arabi

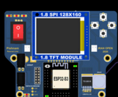
Electronic Specialist

Expriences :

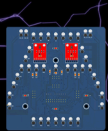
- **IranOpen 2023** — 2nd Place - InSitu
- **IranOpen 2024** — 3rd Place - InSitu
- **IranOpen 2025** — 2nd Place - Rescue Line
- **QFR RCJ 2026** — 1nd Place - Rescue Line

Mohammadamin is responsible for the electronics division of our team. He is highly skilled in working with Altium and demonstrates exceptional proficiency in soldering. This year, he has incorporated significant innovation and creativity into the design and development of our electronic boards. In addition, he worked with Altium Designer to create the boards and collaborated closely in his design and development.

QFR RCJ 2026



The Main Board serves as the heart of the robot, functioning as the central control unit that manages and coordinates all operations. At the core of the Main Board is an ESP-32, which is responsible for overseeing the robot's movements and sensors. The board also houses critical components such as voltage regulators, the main power supply, and other essential circuitry.



The Sensor Board is equipped with 39 pairs of IR sensors for detecting black line strips and two APDS-9960 sensors for identifying green squares. Given the high number of sensors, they require many pins for operation. To optimize pin usage, multiplexers were employed. Specifically, one 4052 multiplexer and two 4067 multiplexers are implemented to efficiently manage the pin count while maintaining full functionality. This design ensures that all sensors can operate effectively while minimizing the overall number of pins required on the board.

This year's connector board has undergone significant changes compared to last year. It is no longer merely a set of auxiliary connectors; it has become one of the robot's main subsystems. This board plays a crucial role, especially in the evacuation zone, where stable operation and reliable connectivity are essential to the robot's overall performance.

The Microswitch Board incorporates a microswitch to detect walls within the evacuation zone, as well as a mechanism for determining whether victims are alive or dead by passing an electrical current. This design ensures accurate and efficient wall detection, along with rapid identification of the victims' status, enhancing the robot's operational efficiency and response capabilities during the mission.

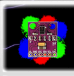
Electronical Innovation

Problem : Separate sensors increased wiring complexity and reduced reliability.

Sloution : A single board combining color, distance, and victim detection.

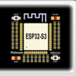
Why It Is Innovative : Unifies multiple sensing functions in one compact module.

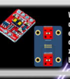
Impact : Improves speed, stability, and detection reliability.



We used 3 APDS-9960 in our robot to detect colors like green and red colors. The APDS-9960 is used for gesture control, proximity sensing, ambientlight detection. Price : 1.5$

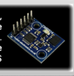
ESP32-S3-WROOM-1 is a compact WiFi and Bluetooth 5 (LE) module powered by a dualcore Xtensa® LX7 processor, designed for IoT and embedded applications with high performance. Price : 3.5$



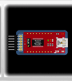


We used a VL53L1X sensor on the VL board, alongside the APDS board, to detect walls and evacuation points in the evacuation zone, and obstacles both inside and outside the evacuation zone, providing accurate distance measurements for better navigation and collision avoidance. Price: 5$

We used a CMPS14 sensor to detect pitch, roll, and yaw, and to accurately detect and navigate ramps. It is also used for the robot's rotations. Although it offers high precision, it is more expensive compared to other options. Price: 30$



We used 43 pairs of IR sensors for efficient line-following, as their affordability and effectiveness make them ideal for this purpose. They are also used to distinguish real live victims from fake ones, to check whether the gripper has successfully grabbed a victim, and for obstacle detection. Price: 0.2$



We designed an uploader converter board for programming the main board, featuring an FT232RL module, C945 transistor, AMS1117, SS56, and several capacitors and resistors to ensure stable communication and reliable uploading (FT232RL module price: 2$).

MECHANIC

AmirFarhan Mohseni

Mechanical Specialist

Expriences :

- **IranOpen 2024** — InSitu
- **IranOpen 2025** — InSitu
- **QFR RCJ 2026** — 1nd Place - Rescue Line

Amirfarhan Mohseni is the sole mechanical specialist of our team. He designed the robot using SolidWorks and managed the 3D printing and assembly processes, ensuring the robot is fully functional and competition-ready. He operates the Bambulab P1S 3D printer and uses Bambu Studio for printing. Amirfarhan is also the author of the Technical Description Paper (TDP), where he applied creativity and introduced several innovative ideas.

QFR RCJ 2026

The body of our robot is fully 3D printed using PLA+ filament, chosen for its balance between rigidity and flexibility. Unlike standard PLA, which is brittle, or TPU and PETG, which are too flexible, PLA+ provides a reliable middle ground for both strength and durability. It also gives the robot a clean and professional appearance. We used eSUN PLA+ filament with the Bambulab P1S printer to ensure consistent quality and precision.

The robot utilizes two types of motors, XC430-W240-T and XL330-M288-T, each assigned to specific functions including locomotion, arm operation, and the victim storage box. The locomotion system is powered by four XC430 motors. The arm mechanism incorporates two XL330 motors for the gripper, along with one XL330 and one XC430 motor for arm movement. The storage box is equipped with one XL330 motor for the holder, which secures the victims to prevent them from falling and assists in separating them, and one XC430 motor for positioning the box.

Mechanical Innovation

Problem: The external arm increased the overall size of the robot and reduced its efficiency.

Solution: We integrated the arm inside the robot to achieve a more compact design.

Why It Is Innovative: Internal arm integration enhances balance while minimizing overall dimensions.

Impact: Improves stability, obstacle handling, and maneuverability.

SCAN ME!

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Designed by Amin Arabi